

PPO for RANOpt AI

Proximal Policy Optimization for AI-assisted 5G/6G RAN optimization

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Simple, thin deck: PPO concepts placed directly in the RANOpt AI context.

Why Reinforcement Learning in RAN?

RL learns by trial and reward; the RAN simulator acts as the environment.

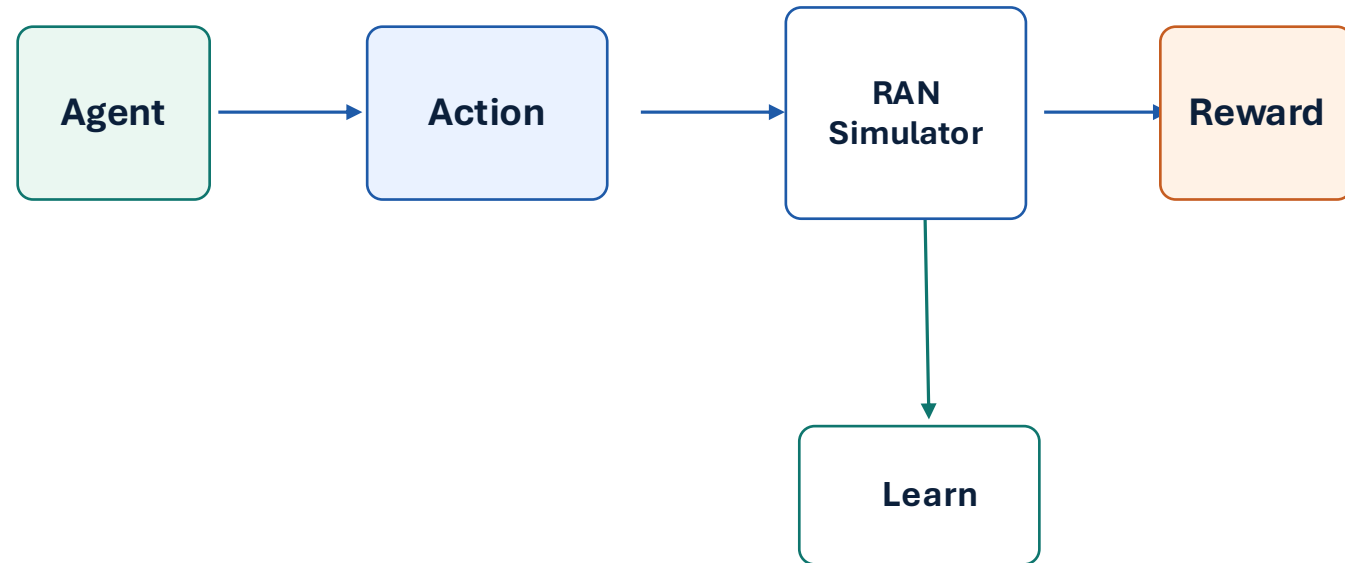
Agent: the AI decision-maker.

Action: scheduler or configuration choice.

Environment: the RAN simulator reacts.

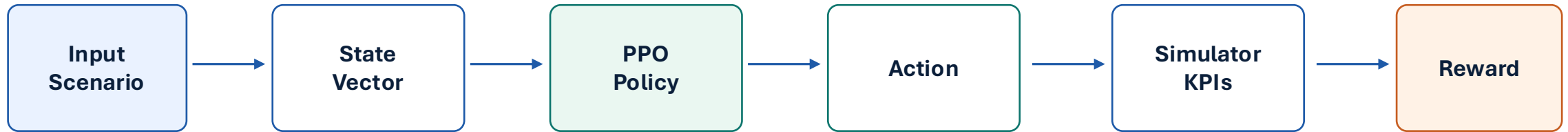
Reward: score from KPIs after the action.

Repeated trials improve the policy over time.



RANOpt AI Environment

The PPO agent observes a RAN state, takes one action, and is scored by simulator KPIs.



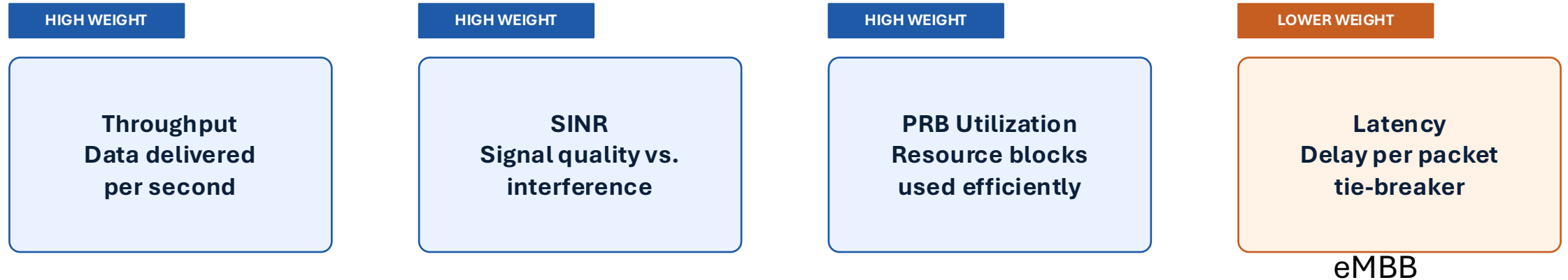
State may include frequency, bandwidth, morphology, active UEs, CQI/SINR, MIMO, CA and scheduler context.

Actions adjust scheduler/configuration choices within allowed limits.

KPIs are normalized and combined into a single reward.

Reward from Four KPIs

For the current eMBB-oriented demo, throughput, SINR and PRB utilization dominate.



$$\text{reward} = w_1 \cdot \text{Throughput} + w_2 \cdot \text{SINR} + w_3 \cdot \text{PRB_Utilization} - w_4 \cdot \text{Latency}$$

All KPIs are normalized to a common scale before combining into one reward number.

The Policy Network

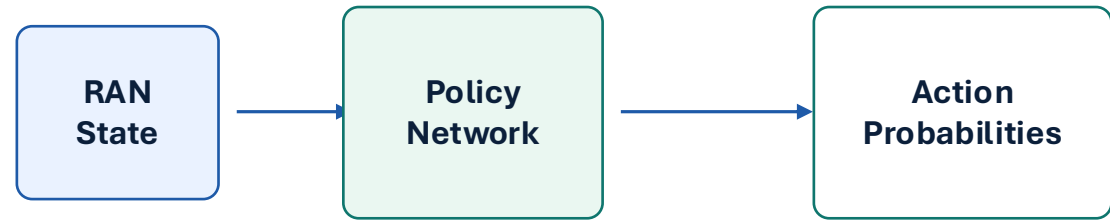
The policy is the agent's playbook: given this RAN state, what should I do?

In PPO, the policy is a neural network.

Input: current network state and KPI context.

Output: probability over possible actions.

Training adjusts the weights so better actions become more likely.

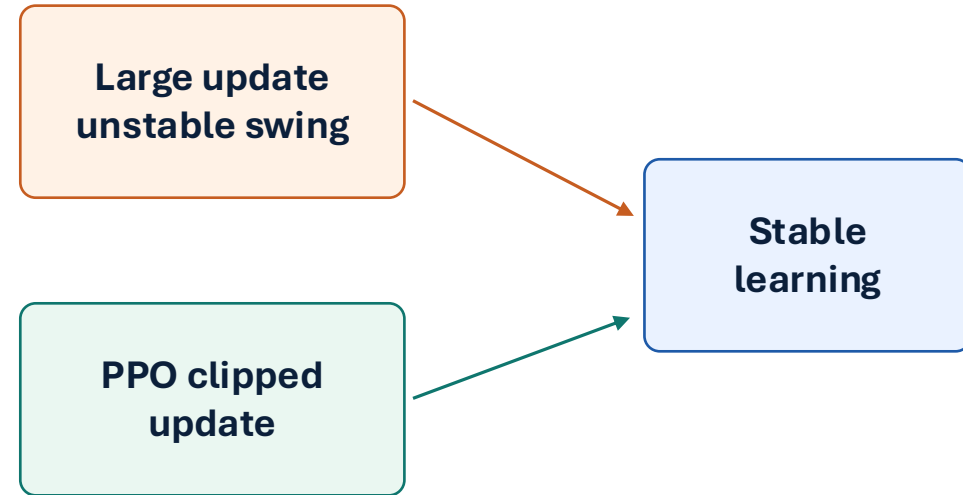


Example: if SINR is low and PRB usage is high, PPO may favor PRB reallocation or scheduler tuning over a less useful action.

Why “Proximal”? Small, Careful Updates

PPO avoids overreacting to one lucky or unlucky outcome.

A reckless update can destroy previously learned behavior.
PPO clips how much the policy can change in one update.
Result: steadier and more reliable improvement.



PPO Training Loop in RANOpt AI

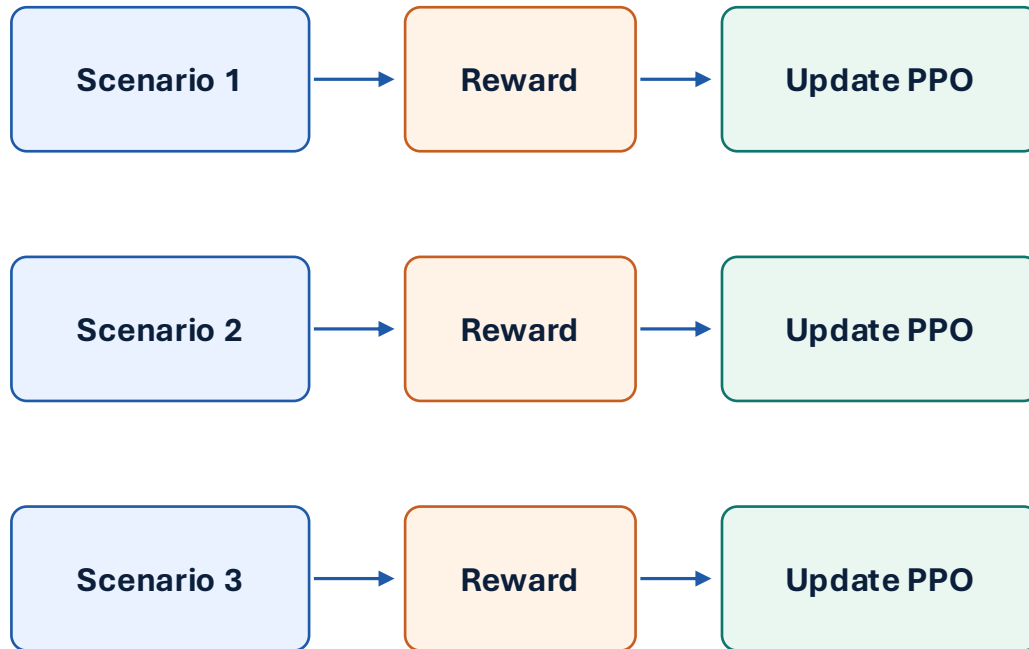
The simulator supplies the reward signal; PPO nudges the policy toward better RAN actions.



Compare: was the result better or worse than expected?
Advantage estimate guides the update direction.
Clipped PPO update prevents unstable policy jumps.

Current RANOpt AI: Contextual Bandit

Each scenario is independent; PPO learns the mapping from state to best immediate action.



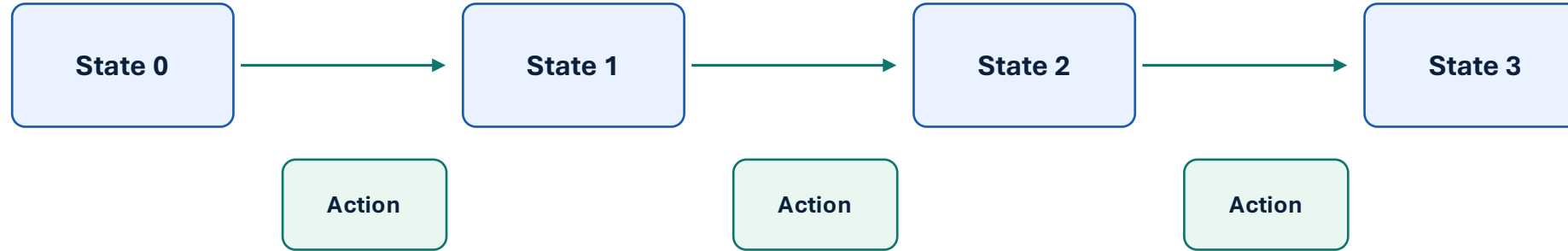
Scenario state resets after every run.

Reward is computed from that one run's KPIs. PPO weights are retained and updated across many runs.

Good for configuration recommendation and scheduler tuning demos.

Future Direction: Multi-Step MDP Digital Twin

A true digital twin would model how today's action changes tomorrow's network state.

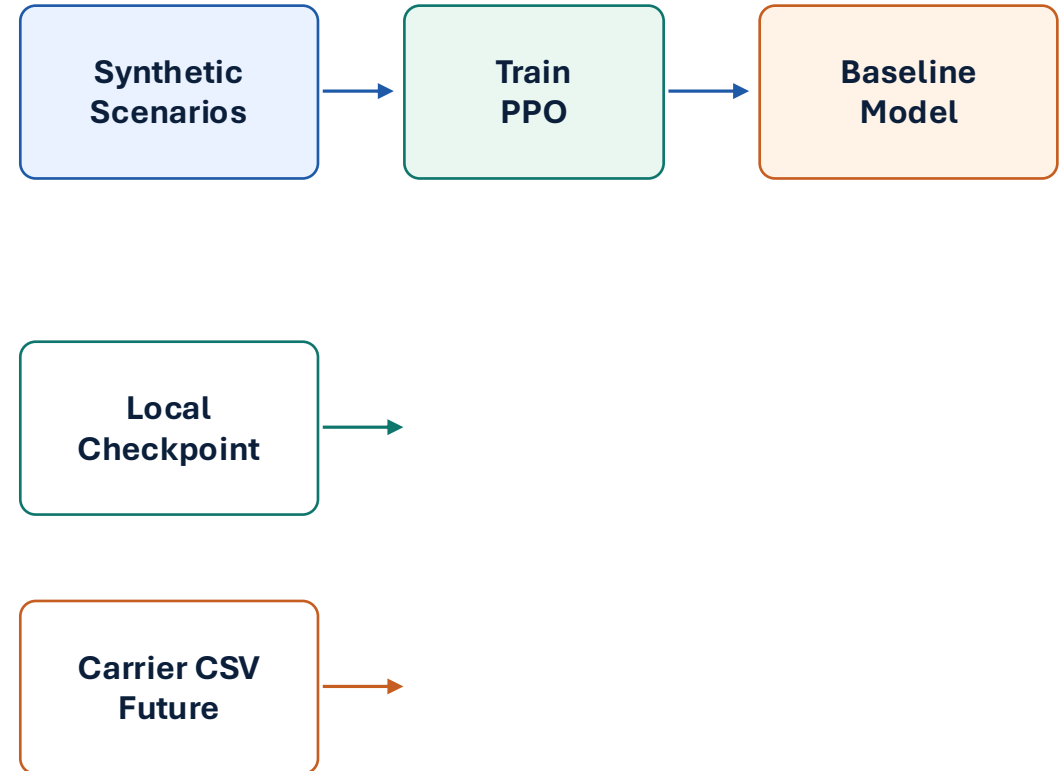


Requires mobility, queue dynamics, traffic arrivals and changing CQI.
Actions influence future states and long-term reward.
More realistic for closed-loop RAN control, but much more complex.
Recommended as a later v18/v19 evolution, not the first v17 step.

PPO Pre-Training Before Live Learning

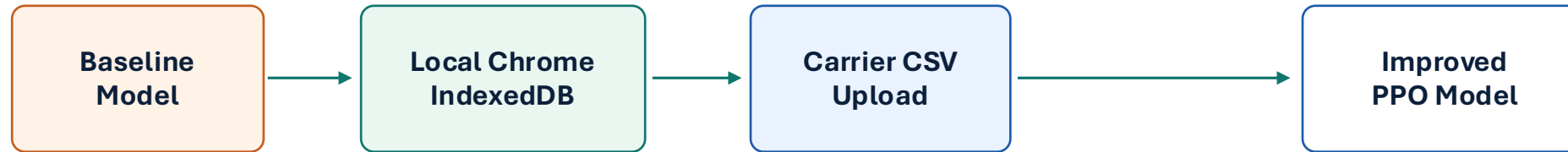
Pre-training starts PPO from a useful baseline rather than from random weights.

Synthetic RANOpt scenarios are the default source.
Previously saved local PPO state can continue learning.
Carrier CSV becomes an additional source when available.
Combined training is preferred for robustness.



Incremental Learning in Chrome

v17 should preserve PPO training across Chrome sessions using IndexedDB and export/import.



Use same Chrome browser, profile, device and RANOpt AI URL.

Store policy/value weights, episode count, reward history and model version.

Add Save, Export, Import and Reset controls.

Do not claim carrier training unless carrier data was actually used.

Summary: PPO in the RANOpt AI Roadmap

A practical path from browser demonstration to operator-refined AI optimization.

PPO learns scheduler/configuration choices from KPI-based rewards.

Current formulation: independent single-step contextual-bandit scenarios.

v17 direction: persistent Chrome training with IndexedDB checkpoints.

Training sources: synthetic scenarios, local history, carrier CSV and combined mode.

Future roadmap: multi-step MDP digital twin for closed-loop RAN control.

